Audio Processing Using Fourier Transforms

Let t denote time in seconds and ω angular frequency in rad/s. The continuous-time Fourier transform and its inverse are

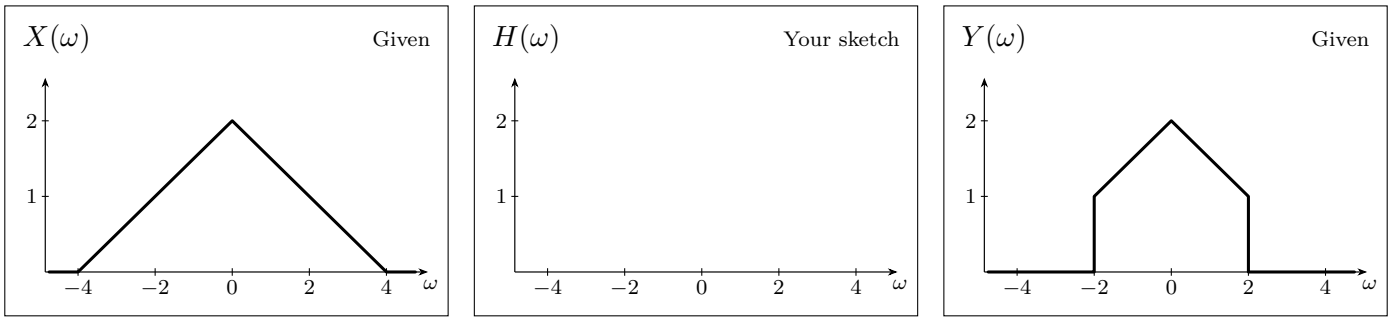
$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt, \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,d\omega.$$

For an LTI system, let $x(t)$ be the input, $h(t)$ its impulse response, and $y(t)$ the output. Write $H(\omega)$ and $Y(\omega)$ for the transforms of $h(t)$ and $y(t)$. Then

$$y(t) = (h * x)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad Y(\omega) = H(\omega)X(\omega).$$

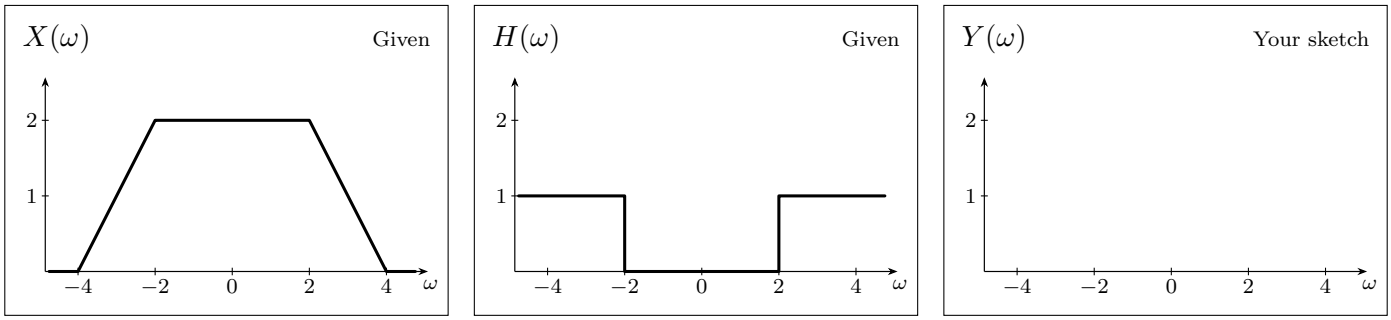
1. Sketch the filter

Sketch $H(\omega)$ for the input and output shown.



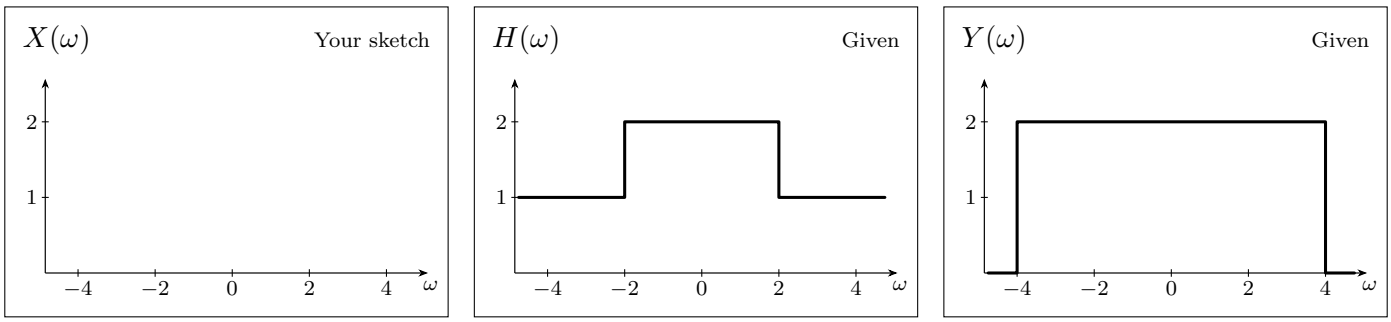
2. Sketch the output

Sketch $Y(\omega)$ for the input and filter shown.



3. Recover the input

Sketch $X(\omega)$ for the filter and output shown.



4. Find the output in the frequency domain

Let $a > 0$. The frequency-domain representations of the input and filter are

$$X(\omega) = \frac{1}{a + j\omega}, \qquad H(\omega) = \begin{cases} 1, & |\omega| < 10 \text{ rad/s}, \\ 0, & |\omega| \geq 10 \text{ rad/s}. \end{cases}$$

Write $Y(\omega)$ as a piecewise function for all ω.

Hands-on Activity: Guitar Fourier Lab

Use the guitar recording in the app. The filter editor uses frequency f in Hz, with $\omega = 2\pi f$.

5. Separate the low and high parts of the clipped guitar

Choose a clipping threshold and a cutoff frequency. Design one filter to hear the sound below the cutoff and another to hear the sound above it.

Clipping threshold _____ % Cutoff frequency _____ Hz

What do you hear below the cutoff?

What do you hear above the cutoff?

6. Single out one note

Set the clipping threshold to 100% and isolate one note (your favorite note in this guitar song file) with a narrow filter (design that filter!).

Fundamental frequency _____ Hz Passband _____ to _____ Hz

When do you hear the note in the filtered recording?